

Xinlin Zhang

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Career Objective

AI/Software Engineer.

Seasoned software engineer with a Master's Degree of AI in AUT. Passionate about real-world AI applications. Seeking to join a team in which I can apply my expertise in building innovative and cutting-the-edge AI systems.

Skills Summary

- proficient in Python, familiar with PyTorch
- **Deep Learning & Computer Vision:** Experienced in designing, implementing, training and evaluating various deep learning models from scratch by using PyTorch, specialized in building customized CNN architectures for image classification and semantic segmentation tasks. Familiar with the architecture of R-CNN family and YOLO series models for object detection.
- **MLOps & Automation:** Capable of automating Label Studio into working flow, integrating SAM2 or self-trained model into Label Studio, realizing interactive model-assistant data labeling or auto-labeling with self-trained models.
- **Large Language Models (LLMs):** Hands-on experience in deploying self-hosted LLMs and deep understanding of prompt engineering.
- **Computer Science Background:** expertise on Computer Science, OS, Network, Algorithm and Database
- **Seasoned Software Engineer:** many years' experience in C/C++, familiar with Linux and Shell
- [Personal Projects Portfolio](https://github.com/xinlin-z/portfolio) (github.com/xinlin-z/portfolio)

Professional Experience

AI Software Research Engineer Intern, TOMRA Food (New Zealand), 11/2025 – Present

- Working in the Data Scientist Team as an Intern.
- Design, Implement, Train and Evaluate highly modernized, customized and configurable deep learning models for multi-class image segmentation tasks. These models have fixed input and output, and need to be run in a memory-constrained real-time environment. The model's performance is good enough to test on the customer's site (waiting for a chance). (Python, PyTorch)
- Automate Label Studio into working flow, realize interactive model-assistant data labelling, and build training and validation data. (Python, Shell)
- Implement the whole working codebase, including models, data augmentation algorithm, training and fine-tuning process, and all other tools. (Python, PyTorch, Shell, Javascript, FastAPI)
- Solve/bypass a few critical issues related to GPU VRAM and onnx runtime (TensorRT).

Senior System Software Engineer, Astute-Tec (China), 2021 – 2023

- Designed and implemented a highly optimized and configurable BGRA image comparison algorithm and a hybrid H.264/JPEG transportation protocol, reducing CPU utilization from over 20% to below 5% in a cloud desktop system during static or minimal screen activity. (C++, Python)
- Designed and implemented a priority-based multi-TCP schedule algorithm, significantly improving user responsiveness and fluidity in a highly loaded cloud virtual desktop environment. (C++)
- Recipient of the “Network and Video Codec Expert Award”, May 2022.

Software Engineer, VP, Co-founder, GWorld (China), 2014 – 2021

- Solely designed and implemented an auto-production system for manufacturing company proprietary optical modules ($\leq 10G$) and OEM modules. (Python, SQLite)

Software Engineer, ZTE R&D Center (China), 2006 – 2013

- Developed and maintained many features in the TCP/IP stack (such as Ethernet, VLAN, Routing Protocol, ACL, and RTP) which were integrated into many different telecommunication products. (C)
- Recipient of the Department's “Best Programmer Award”, December 2009.

Education

Auckland University of Technology (New Zealand), Feb. 2025 – Present (graduate in July 2026)

Master of Computer and Information Science (courses, dissertation and internship are all about AI)

Coursera.org:

AI Agent Developer (Vanderbilt University), [Certificate](#) earned in Jun. 2025

Mathematics for Machine Learning and Data Science (DeepLearning.AI), [Certificate](#) earned in Feb. 2025

Northeastern University (China), Sep. 1998 – Jul. 2002

Bachelor of Engineering Degree in Management Information Systems (under the department of Computer Science)

Interests

Reading, Coding, Hiking and Green Tea.

Reference

Patrick Lindegger

Data Scientist in TOMRA Food

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